Lecture 7

States, Liouville's theorem and the second answer

Objectives. *Lectures 5 and 6 described the evolution of observables along Hamiltonian flows. This lecture describes the evolution of states. A state need not be a point. Incomplete knowledge is represented by a probability distribution on phase space. Transporting that distribution requires knowing how the flow changes volume. Liouville's theorem says that Hamiltonian flows preserve volume. Smooth densities then satisfy the Liouville equation. Evolving states or evolving observables gives the same predictions. Volume also settles the question Lecture 1 asked about attractors. The lecture closes with the second answer to the course's question: a classical mechanical system is a Poisson algebra of observables, a distinguished element H, and a convex set of states.*

By the end of this lecture you will be able to:

- Calculate expectations in states formed from densities and point masses.
- Derive volume preservation for Hamiltonian flows.
- Transport a density using the Liouville equation.
- Explain the agreement of the state and observable pictures.

7.1 States

Throughout, $\mathcal{M} \subseteq \mathbb{R}^{2n}$ is open with coordinates $x = (q, p)$, observables are the real C^∞ functions on $\mathcal{M}$ (Definition 5.1), and $H \in C^\infty(\mathcal{M})$ has flow φ_t, the flow of $\dot{x} = J\nabla H$ (4.9). Integrals are ordinary multiple integrals with $\mathrm{d}x = \mathrm{d}q_1 \cdots \mathrm{d}q_n \, \mathrm{d}p_1 \cdots \mathrm{d}p_n$, and vol is the volume they give.

Until now a state has been a point of $\mathcal{M}$ (Definition 1.1), specifying the initial data that determine the motion. An experimenter rarely knows those data exactly. What a repeatable preparation delivers is a *distribution* of initial data, and every prediction is then a distribution of readings.

Definition 7.1 (Density, state)**.** A *density* on $\mathcal{M}$ is a continuous $\rho \colon \mathcal{M} \to [0, \infty)$ with $\int_{\mathcal{M}} \rho(x)\,\mathrm{d}x = 1$. A *state* is a finite convex combination

$$\sigma = \sum_{k=1}^{r} \lambda_k \rho_k + \sum_{j=1}^{s} \mu_j \delta_{z_j}, \qquad \lambda_k, \mu_j \geq 0, \quad \sum_k \lambda_k + \sum_j \mu_j = 1, \tag{7.1}$$

of densities ρ_k and of points $z_j \in \mathcal{M}$. It acts on observables by

$$\langle f \rangle_\sigma = \sum_k \lambda_k \int_{\mathcal{M}} f \rho_k \, \mathrm{d}x + \sum_j \mu_j f(z_j), \tag{7.2}$$

the *expectation* of f, whenever the integrals converge absolutely. The states form a set $\mathcal{S}(\mathcal{M})$; a state of the form δ_z is a *Dirac state*.

Every bounded observable has an expectation. For a growing f the expectation may be infinite, and the state then predicts no mean for that device. The functional (7.2) is linear, positive and normalised, and $\mathcal{S}(\mathcal{M})$ is *convex* by construction: a combination of two sums of the form (7.1) is another such sum (Figure 7.1).

Example 7.2 (A mixed state). On $\mathcal{M} = \mathbb{R}^2$ let $\rho(q,p) = e^{-(q^2+p^2)/2}/2\pi$. This is a density, since polar coordinates give $\int \rho \, \mathrm{d}x = \int_0^\infty r e^{-r^2/2} \, \mathrm{d}r = 1$. Let $\sigma = \frac{1}{2}\rho + \frac{1}{2}\delta_{(1,0)}$, half the density and half the point $(1,0)$. Take the observable $f = \frac{1}{2}(q^2+p^2)$. Polar coordinates give

$$\int f\rho \, \mathrm{d}x = \int_0^\infty \frac{1}{2} r^3 e^{-r^2/2} \, \mathrm{d}r = 1, \qquad \text{and} \qquad f(1,0) = \frac{1}{2},$$

so $\langle f \rangle_\sigma = \frac{1}{2}\cdot 1 + \frac{1}{2}\cdot\frac{1}{2} = \frac{3}{4}$. For the observable q, the density ρ is even in q, so $\int q\rho \, \mathrm{d}x = 0$, and $\langle q \rangle_\sigma = \frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 1 = \frac{1}{2}$.

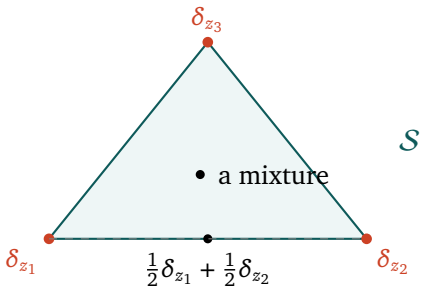


Figure 7.1: The mixtures of three Dirac states form a triangle: the vertices are the Dirac states and every other point is a mixture. Proposition 7.3 characterises the Dirac states on any $\mathcal{M}$.

The reading of an observable

A state predicts more than the number $\langle f \rangle_\sigma$: the probability that a reading of f falls below λ is $\langle \mathbf{1}_{\{f<\lambda\}} \rangle_\sigma$, and letting λ run gives the whole distribution of the reading (Figure 7.2). Indicators jump, so they are not observables — Lecture 5 set them aside for exactly this moment — but a state integrates them anyway, and $\langle \mathbf{1}_B \rangle_\sigma$ is the probability of finding the system in the region B. In Example 7.2, the probability of $q > 0$ is $\frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot 1 = \frac{3}{4}$.

Pure states

Proposition 7.3 (Pure states). *For $\sigma \in \mathcal{S}(\mathcal{M})$ the following are equivalent.*

1. *$\sigma = \delta_z$ for some $z \in \mathcal{M}$.*
2. *Each of the $2n$ coordinates $x_a = q_1, \dots, p_n$ has $\langle x_a^2 \rangle_\sigma < \infty$ and variance $\mathrm{Var}_\sigma(x_a) =$*

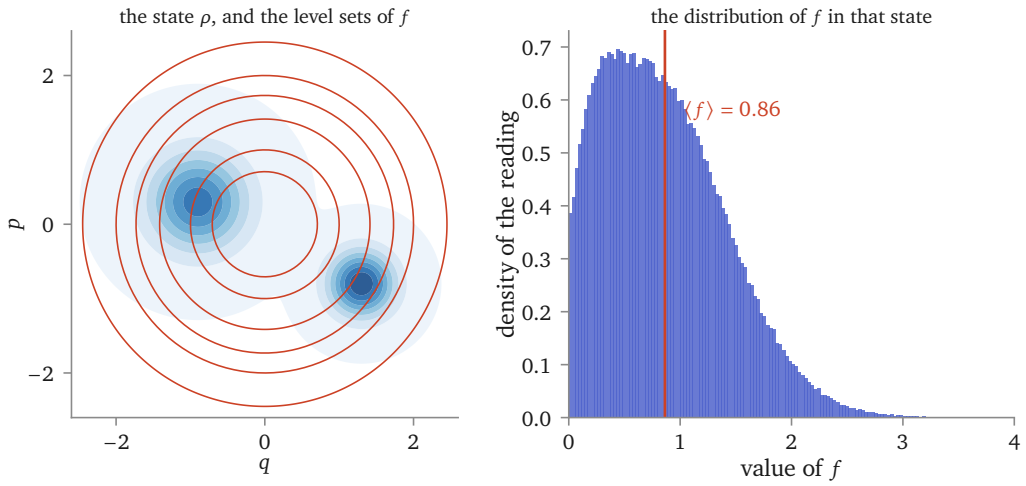


Figure 7.2: Left: a state (a mixture of two Gaussians) with the level sets of $f = \frac{1}{2}(q^2 + p^2)$ over it. Right: the distribution of the reading of f in that state, with $\langle f \rangle$ marked.

$\langle x_a^2 \rangle_\sigma - \langle x_a \rangle_\sigma^2 = 0$. *Such a state is called* dispersion-free.

3. *σ is an extreme point of $\mathcal{S}(\mathcal{M})$: it is not $\alpha\sigma_1 + (1-\alpha)\sigma_2$ with $0 < \alpha < 1$ and $\sigma_1 \neq \sigma_2$.*

Such a state is called pure*, and every other state is* mixed.

Proof. One observation is used twice. $(*)$ *If $g \geq 0$ is continuous on $\mathcal{M}$, $g(x) > 0$ for $x \neq c$, and $\langle g \rangle_\sigma = 0$, then $c \in \mathcal{M}$ and $\sigma = \delta_c$.* Indeed every term of (7.2) applied to g vanishes. A density component would give $\int \rho_k g = 0$ with a non-negative continuous integrand, so ρ_k would vanish off the point c and, by continuity, everywhere — against $\int \rho_k = 1$. So every $\lambda_k = 0$, some $\mu_j > 0$, and $\mu_j g(z_j) = 0$ leaves $z_j = c$ whenever $\mu_j > 0$.

(1)⇒(2). We have $\langle f^2 \rangle_{\delta_z} = f(z)^2 = \langle f \rangle_{\delta_z}^2$.

(2)⇒(1). Each $\langle x_a \rangle_\sigma = c_a$ is finite, since $|x_a| \leq 1 + x_a^2$. Then $\langle |x - c|^2 \rangle_\sigma = \sum_a \mathrm{Var}_\sigma(x_a) = 0$, and $(*)$ with $g = |x - c|^2$ gives $\sigma = \delta_c$.

(1)⇒(3). If $\delta_z = \alpha\sigma_1 + (1-\alpha)\sigma_2$ with $0 < \alpha < 1$, take the bounded observable $g = |x - z|^2 / (1 + |x - z|^2)$. Then $0 = g(z) = \alpha\langle g \rangle_{\sigma_1} + (1-\alpha)\langle g \rangle_{\sigma_2}$ with both expectations non-negative, so both vanish, and $(*)$ gives $\sigma_1 = \sigma_2 = \delta_z$.

(3)⇒(1). The idea is that a density can be split in two by multiplying it by a function with values between 0 and 1. Let σ be no Dirac state, written as in (7.1) with positive weights. If some $\mu_j \in (0, 1)$ then $\sigma = \mu_j \delta_{z_j} + (1 - \mu_j)\sigma'$, and $\sigma' \neq \delta_{z_j}$ because otherwise $\sigma = \delta_{z_j}$: a non-trivial mixture. No μ_j is 1, since that is the excluded case. So every $\mu_j = 0$ and $\sigma = \sum_k \lambda_k \rho_k$ is itself a density ρ, a convex combination of densities being one. Put $\chi = e^{q_1}/(1 + e^{q_1})$, continuous with $0 < \chi < 1$, and $\alpha = \int \chi\rho \in (0, 1)$; then $\rho = \alpha\rho_1 + (1-\alpha)\rho_2$ with $\rho_1 = \chi\rho/\alpha$ and $\rho_2 = (1-\chi)\rho/(1-\alpha)$, both densities. They differ: $\rho_1 = \rho_2$ would force $\chi \equiv \alpha$ on the non-empty open set $\{\rho > 0\}$, which a strictly increasing function of q_1 is not. ∎

A pure state is thus a point of $\mathcal{M}$ with three equivalent descriptions. It specifies exact initial data, gives a definite value to every position and momentum coordinate, and cannot be prepared as a nontrivial mixture of two other states.

7.2 Liouville's theorem

To transport states we must first know what the flow does to volume. Lecture 6 answered this through the symplectic group: $D\varphi_t \in \mathrm{Sp}(2n)$ and $\det D\varphi_t = \pm 1$ by Proposition 6.14, hence $+1$, since it is continuous in t and equals 1 at $t = 0$. The proof below needs no symplectic linear algebra and applies to any smooth vector field. This lets us compare a Hamiltonian flow with the damped pendulum's flow.

Lemma 7.4 (Abel–Liouville formula). *Let $A : I \to \mathbb{R}^{N\times N}$ be continuous and let $M : I \to \mathbb{R}^{N\times N}$ solve $\dot M = A(t)M$. Then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\det M(t) = \operatorname{tr} A(t)\,\det M(t), \qquad \text{so} \qquad \det M(t) = \det M(t_0)\exp\int_{t_0}^{t}\operatorname{tr} A(\tau)\,\mathrm{d}\tau. \tag{7.3}$$

Proof. The determinant is multilinear in the columns $m_1, \dots, m_N$ of M, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\det M = \sum_{j=1}^{N}\det\bigl[m_1, \dots, \dot m_j, \dots, m_N\bigr] = \sum_{j=1}^{N}\det\bigl[m_1, \dots, Am_j, \dots, m_N\bigr].$$

The last sum equals $\operatorname{tr} A \cdot \det M$. Indeed, when M is invertible write $Am_j = \sum_i B_{ij}m_i$ with $B = M^{-1}AM$; only the $i = j$ term survives in the j-th determinant, which is therefore $B_{jj}\det M$, and $\sum_j B_{jj} = \operatorname{tr} B = \operatorname{tr} A$. Both sides are polynomials in the entries of M and A, and they agree on the invertible matrices, which are dense, so they agree everywhere. Finally $\mathrm{d}/\mathrm{d}t\bigl(\det M\, e^{-\int \operatorname{tr} A}\bigr) = 0$. ∎

Theorem 7.5 (Liouville's theorem · *proved modulo T2, T5 (cited)*). *Let X be a C^∞ vector field on $\mathcal{M}$ with flow φ_t. Then*

$$\det D\varphi_t(x) = \exp\int_0^t \operatorname{div} X\bigl(\varphi_\tau x\bigr)\,\mathrm{d}\tau \tag{7.4}$$

wherever $\varphi_t x$ is defined. In particular, if $\operatorname{div} X \equiv 0$ then $\det D\varphi_t \equiv 1$ and $\operatorname{vol}\bigl(\varphi_t(D)\bigr) = \operatorname{vol}(D)$ for every bounded region D with $\overline{D} \subseteq \mathcal{M}$ on which φ_t is defined. For $X = X_H = J\nabla H$ this hypothesis holds:

$$\operatorname{div} X_H = \sum_{i=1}^{n}\Bigl(\frac{\partial}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial}{\partial p_i}\frac{\partial H}{\partial q_i}\Bigr) = 0, \tag{7.5}$$

by equality of mixed partial derivatives.

Proof. Fix x and put $M(t) = D\varphi_t(x)$. By T5 (Theorem 1.8) M is C^1 in t on the interval I_x where the motion exists (Theorem 1.6) and solves the variational equation $\dot M = DX(\varphi_t x)M$ with $M(0) = I$. The trace of its coefficient matrix is $\operatorname{tr} DX = \operatorname{div} X$, so (7.3) gives (7.4). When $\operatorname{div} X \equiv 0$ the exponent vanishes, and the change of variables $y = \varphi_t(x)$ gives $\operatorname{vol}(\varphi_t(D)) = \int_D \bigl|\det D\varphi_t(x)\bigr|\,\mathrm{d}x = \operatorname{vol}(D)$. ∎

In Figure 7.3, a region of initial conditions is stretched into a thin filament. At a fixed resolution it appears to spread out, although its area is unchanged. Each motion conserves its energy, so none crosses the separatrix. The librating and rotating parts remain separate.

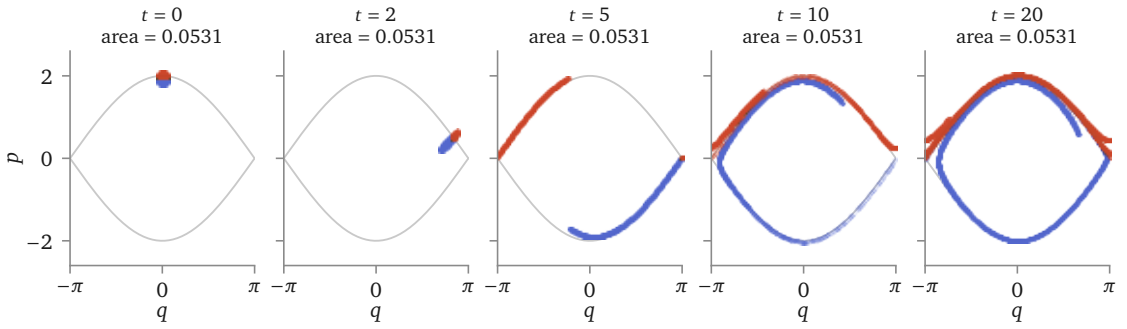


Figure 7.3: A disc of initial conditions straddling the pendulum's separatrix $H = 2$, drawn on the cylinder at $t = 0, 2, 5, 10, 20$; blue librates, red rotates. The shape changes while the area is preserved. The number under each panel is the area of the transported boundary, computed with a numerical method that does not itself preserve area, so that its constancy is a measurement.

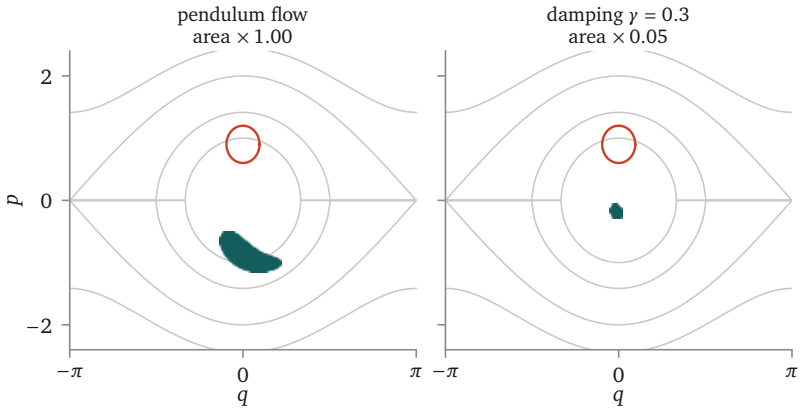


Figure 7.4: A disc of initial conditions inside the separatrix after $t = 10$: under the pendulum's flow, and under the damped pendulum with $\gamma = 0.3$; the red circle is where it started. The damped field has $\operatorname{div} X = -\gamma$, so by (7.4) the disc's area is multiplied by e^{-3}.

The contrast: friction

For the damped pendulum (1.16), written as $\dot q = p$, $\dot p = -\sin q - \gamma p$, the divergence is $-\gamma$, so (7.4) gives $\det D\varphi_t = e^{-\gamma t}$ exactly: every volume shrinks at the same exponential rate, whatever the orbits do (Figure 7.4). This is the quantitative form of the contrast drawn in § 1.6, and § 7.4 turns it into a theorem.

7.3 The two pictures

From here on H does not depend on t and its flow is complete (Definition 1.9), as for $\frac{1}{2}|p|^2 + V(q)$ with $V \in C^\infty(\mathbb{R}^n)$ bounded below (Proposition 1.10).

Lecture 5 evolved the observables while keeping the state fixed: $f_t = f \circ \varphi_t$ and $\partial_t f_t = \{f_t, H\}$ (Theorem A, Theorem 5.15). That is the *Heisenberg picture*. The alternative is to keep the observables fixed and evolve the state.

Transporting a state

A Dirac state follows its phase-space point. After time t the system prepared at z is at $\varphi_t z$, so δ_z becomes $\delta_{\varphi_t z}$. For a density the rule must be found. The probability of finding the system in a region D at time t is the probability that it started in $\varphi_{-t}(D)$, so the transported density ρ_t has to satisfy $\int_D \rho_t = \int_{\varphi_{-t}(D)} \rho$ for every D. Change variables by $x = \varphi_{-t}(y)$, whose Jacobian is 1 by Theorem 7.5:

$$\int_{\varphi_{-t}(D)} \rho(x)\,\mathrm{d}x = \int_D \rho\big(\varphi_{-t}(y)\big)\mathrm{d}y.$$

As this holds for every D and both integrands are continuous,

$$\rho_t = \rho \circ \varphi_{-t}. \tag{7.6}$$

Volume preservation is needed here. Without it the right-hand side would carry a Jacobian factor, and (7.6) would be false. Transporting each component of (7.1) this way gives σ_t, whose densities are again continuous, non-negative and of integral 1: the transported state is again a state.

The sign in (7.6) has a reason. Observables are read where they are, so they are composed with φ_t; a density is *carried along*, so its value at a point now is its old value at the point that flowed *into* it, and it is composed with φ_{-t}.

Theorem 7.6 (The Liouville equation · *proved modulo T2, T5 (cited)*)**.** *Let ρ be a density of class C^∞. Then $(t,x) \mapsto \rho_t(x)$ is C^∞ and*

$$\frac{\partial \rho_t}{\partial t} = \{H, \rho_t\}\,. \tag{7.7}$$

Proof. Put $u_s = \rho \circ \varphi_s$. By Theorem A it is C^∞ in (s,x) with $\partial_s u_s = \{u_s, H\}$. Since $\rho_t = u_{-t}$, the chain rule gives $\partial_t \rho_t = -(\partial_s u_s)\big|_{s=-t} = -\{\rho_t, H\} = \{H, \rho_t\}$. ∎

Thus the evolution of observables and of smooth densities uses the two orders of the same bracket. The operators $\{\cdot, H\}$ and $\{H, \cdot\}$ are negatives of one another.

Which C^∞ densities are stationary? If $\rho_t = \rho$ for every t then (7.7) gives

$$\{H, \rho\} = 0; \tag{7.8}$$

conversely (7.8) makes ρ conserved (Proposition 5.11), constant along every motion, so $\rho \circ \varphi_{-t} = \rho$. The stationary densities are the conserved observables that happen to be densities. For example, let g be a smooth function with $g(H) \geq 0$ and $0 < \int g(H)\,\mathrm{d}x < \infty$. Then $g(H)$, divided by that integral, is a stationary density.

The two pictures must make the same predictions.

Theorem 7.7 (The pictures agree · *proved modulo T2, T5 (cited)*)**.** *Let σ be a state and $t \in \mathbb{R}$. Then for every observable f with $\langle |f| \rangle_{\sigma_t} < \infty$ — equivalently, by the same change of variables, $\langle |f \circ \varphi_t| \rangle_\sigma < \infty$ —*

$$\langle f \circ \varphi_t \rangle_\sigma = \langle f \rangle_{\sigma_t}. \tag{7.9}$$

Proof. For a Dirac component both sides are $f(\varphi_t z)$. For a density component, change variables by $y = \varphi_t(x)$, whose Jacobian is 1:

$$\int f\big(\varphi_t x\big)\rho(x)\,\mathrm{d}x = \int f(y)\,\rho\big(\varphi_{-t} y\big)\mathrm{d}y = \int f \rho_t\,\mathrm{d}y.$$ ∎

For example, take the oscillator with $m = \nu = 1$ and the observable $f = q$, in a state with $\langle q\rangle_\sigma = 1.2$ and $\langle p\rangle_\sigma = 0$ (Figure 7.5). By Example 5.16, $f \circ \varphi_t = q\cos t + p\sin t$. Linearity of the expectation gives

$$\langle f \circ \varphi_t\rangle_\sigma = \langle q\rangle_\sigma \cos t + \langle p\rangle_\sigma \sin t = 1.2\cos t.$$

The state picture gives the same number. Let σ be a round Gaussian density centred at $(1.2, 0)$. The flow is a rotation, which preserves distances, so (7.6) gives the same round Gaussian centred at $\varphi_t(1.2, 0) = (1.2\cos t, -1.2\sin t)$. Its mean position is $\langle q\rangle_{\sigma_t} = 1.2\cos t$.

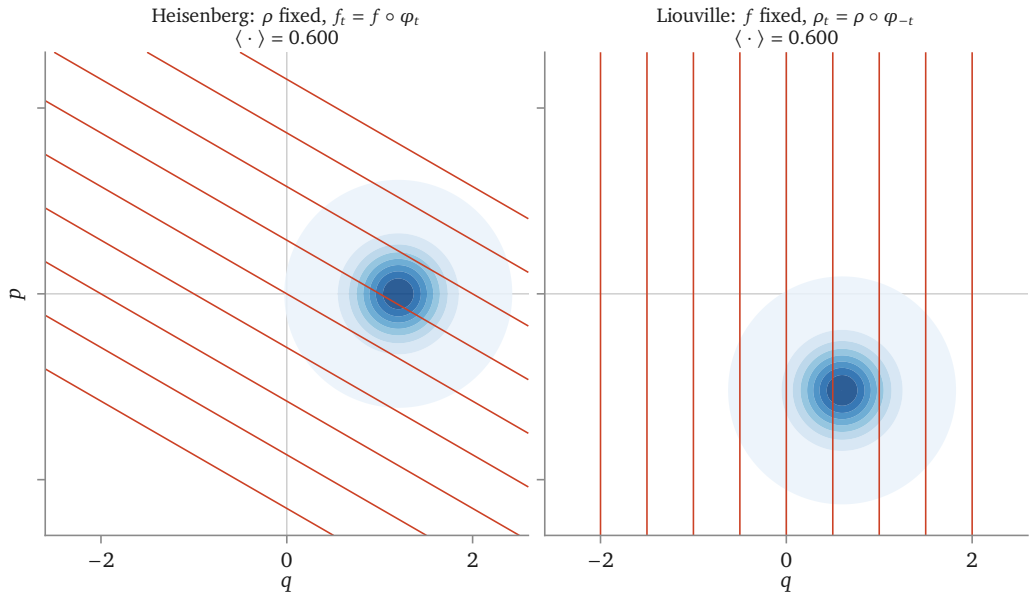


Figure 7.5: The two pictures of one number, for the oscillator and $f = q$, in a Gaussian state centred at $(1.2, 0)$. Left: the state stays and the level sets of $f_t = f \circ \varphi_t$ turn. Right: the level sets of f stay and the state turns the other way. The expectation is the same, and it is $1.2\cos t$. The panels are drawn at $t = \pi/3$.

Adjointness

There is a second, differential way to see (7.9).

Proposition 7.8 (The bracket is a divergence; the integral is cyclic)**.** *For $f, g, h \in C^\infty(\mathcal{M})$,*

$$\{f, g\} = \nabla f \cdot J\nabla g = \operatorname{div}\big(f\, X_g\big). \tag{7.10}$$

If moreover fh has compact support in $\mathcal{M}$, that is, it vanishes outside a closed bounded subset of $\mathcal{M}$, then

$$\int_{\mathcal{M}} \{f, g\}\, h \,\mathrm{d}x = \int_{\mathcal{M}} f\, \{g, h\} \,\mathrm{d}x. \tag{7.11}$$

Proof. We have $\nabla f \cdot J\nabla g = \sum_i (\partial_{q_i} f\, \partial_{p_i} g - \partial_{p_i} f\, \partial_{q_i} g) = \{f, g\}$, and $\operatorname{div}(fX_g) = \nabla f \cdot X_g + f \operatorname{div} X_g = \{f, g\}$ by (7.5). For a compactly supported a the field aX_g extends by zero to a C^∞ field on $\mathbb{R}^{2n}$ of compact support, and each term $\int \partial_b (aX_g)_b$ vanishes on integrating that one variable first: $\int \{a, g\}\,\mathrm{d}x = \int \operatorname{div}(aX_g)\mathrm{d}x = 0$, with no boundary regularity of $\mathcal{M}$ needed. Apply this

to $a = fh$ and use Leibniz (Proposition 5.3): $0 = \int \{fh, g\} = \int \{f, g\}\ h + \int f\ \{h, g\}$, which is (7.11). ∎

Some decay is genuinely needed: for $f = q$, $g = p$, $h = 1$ on $\mathcal{M} = \mathbb{R}^2$ the left side of (7.11) is $\int 1\, \mathrm{d}x = \infty$ and the right side is 0. Under the compact-support assumption, differentiate the two sides of (7.9) at $t = 0$, for a compactly supported f and a C^∞ density ρ. The left gives $\int \{f, H\}\ \rho$ by Theorem A; the right gives $\int f\, \partial_t \rho_t|_{t=0} = \int f\ \{H, \rho\}$ by (7.7). That the two agree is precisely (7.11) with $g = H$ and $h = \rho$.

In words: with respect to the pairing $\langle \rho, f \rangle = \int \rho f$, the generator $\{\cdot, H\}$ of the Heisenberg equation and the generator $\{H, \cdot\}$ of the Liouville equation are *adjoint* to one another. So the Liouville equation is the adjoint of the Heisenberg equation. Lecture 12 lists the quantum counterparts of both equations, and nothing here depends on that list.

7.4 No attractors

Lecture 1 asked what forbids attractors in mechanics. Volume settles the question, for attractors of every kind.

Definition 7.9 (Trapping region, attractor)**.** A *trapping region* for a flow is a non-empty bounded open $U \subseteq \mathcal{M}$ with $\overline{U} \subseteq \mathcal{M}$ on which the flow is defined for all $t \geq 0$ and satisfies $\varphi_t(U) \subseteq U$. The set it traps is $\Lambda = \bigcap_{t \geq 0} \varphi_t(U)$, and an *attractor* is a set of volume zero trapped by some trapping region.

Sinks and attracting limit cycles are attractors in this sense.

Theorem 7.10 (No attractors of volume zero · *proved modulo T2, T5 (cited)*)**.** *Let U be a trapping region for X_H. Then* $\mathrm{vol}(\Lambda) = \mathrm{vol}(U) > 0$. *In particular a Hamiltonian flow has no attractor.*

Proof. For $t \geq s \geq 0$ we have $\varphi_t(U) = \varphi_s(\varphi_{t-s}(U)) \subseteq \varphi_s(U)$, so the open sets $\varphi_t(U)$ decrease as t grows, and Λ is already the intersection over the integers $m \geq 0$. Each $\varphi_m(U)$ is a subset of U with $\mathrm{vol}(\varphi_m(U)) = \mathrm{vol}(U) < \infty$ by Theorem 7.5, so $\mathrm{vol}\big(U \setminus \varphi_m(U)\big) = 0$; and $U \setminus \Lambda = \bigcup_{m \geq 0} \big(U \setminus \varphi_m(U)\big)$ is a countable union of sets of volume zero, hence of volume zero.[1] So $\mathrm{vol}(\Lambda) = \mathrm{vol}(U)$, which is positive because U is open and non-empty. ∎

The trapping-region hypothesis is essential. The flow of $H = -qp$ is $\varphi_t(q, p) = (e^{-t}q, e^t p)$, which preserves area while every point approaches the null line $\{q = 0\}$. There is no trapping region: the orbits run off along p as they close in. Volume stops a flow compressing a region onto a thin set, not stretching one out along one.

This closes the question of § 1.6. The damped pendulum has an attractor because its flow contracts volume at the rate γ; a Hamiltonian flow contracts nothing, so Theorem 7.10 rules out the sink, the attracting limit cycle and every other attractor of volume zero. What a Hamiltonian flow can do instead is Figure 7.3: spread a blob out without shrinking it.

7.5 What a mechanical system is: the second answer

The question of Lecture 1 now has its second answer.

[1] A countable union of sets of volume zero has volume zero, by covering the m-th by open sets of total volume $\varepsilon 2^{-m}$.

Definition 7.11 (Classical mechanical system, second answer)**.** A *classical mechanical system* consists of

1. a Poisson algebra $\mathcal{A}$ of observables (Definition 5.7);
2. a distinguished element $H \in \mathcal{A}$, the Hamiltonian;
3. a convex set $\mathcal{S}$ of states. A state assigns to an observable its expectation $\langle f \rangle_\sigma$, linearly and positively, with $\langle 1 \rangle_\sigma = 1$, on the observables it can read. The extreme points of $\mathcal{S}$ are the pure states.

The dynamics is generated by H: observables evolve by $\dot{f} = \{f, H\}$ and states by the adjoint equation, which must carry $\mathcal{S}$ into itself, and the two give the same expectations, $\langle f_t \rangle_\sigma = \langle f \rangle_{\sigma_t}$.

The clause "on the observables it can read" matters. For the states of Definition 7.1, the observables that a state can read are not all of $\mathcal{A}$: the density ρ of Example 7.2 gives no value to $e^{q^2+p^2}$, since the integral diverges.

Each clause has been proved for $\mathcal{A} = C^\infty(\mathcal{M})$ and the states of Definition 7.1: the algebra is a Poisson algebra (Theorem 5.5 and Proposition 5.3); the flow acts on it by automorphisms (Theorem B, Theorem 6.4); the states are convex with the Dirac states as extreme points (Proposition 7.3); for a complete flow the transported state is again a state (§ 7.3); and the two pictures agree (Theorem 7.7). The pendulum passes through all three layers:

	Lagrange	**Hamilton**	**Poisson**
Arena	$Q = S^1$	$\mathbb{R}^2$, the angle read modulo 2π (§ 1.5)	$\mathcal{A} = C^\infty(\mathcal{M})$ with $\{\cdot,\cdot\}$
Object	the curve $q(t)$	the point (q, p)	the state σ
Law	for $\mathcal{L} = \frac{1}{2}\dot{q}^2 + \cos q - 1$, the action is stationary with fixed ends	$\dot{x} = J\nabla H$, $H = \frac{1}{2}p^2 + 1 - \cos q$	$\dot{f} = \{f, H\}$; $\partial_t \rho = \{H, \rho\}$ for smooth densities; points follow the flow
Reading	—	$f(q, p)$	$\langle f \rangle_\sigma$

The three columns describe the same pendulum. The Lagrange description emphasises curves and coordinate invariance. The Hamilton description uses points in phase space, and the Poisson description also allows statistical states. Conversely, Proposition 7.3 identifies the Dirac states with points, and Theorem 4.5 identifies their orbits with curves in configuration space.

Summary.

- *Handed on:* states, finite mixtures of densities and points, with their expectations (Definition 7.1); pure = Dirac = dispersion-free = extreme (Proposition 7.3); Liouville's theorem, $\det D\varphi_t = \exp \int_0^t \operatorname{div} X$ and $\operatorname{div} X_H = 0$ (Theorem 7.5); densities move by $\rho_t = \rho \circ \varphi_{-t}$, and smooth densities satisfy $\partial_t \rho_t = \{H, \rho_t\}$ (Theorem 7.6); the two pictures agree (Theorem 7.7), and the Liouville equation is the adjoint of the Heisenberg equation (Proposition 7.8); a Hamiltonian flow has no attractor (Theorem 7.10), which answers the question of § 1.6.
- *The second answer:* a classical mechanical system is a Poisson algebra of observables, a distinguished H, and a convex set of states (Definition 7.11).
- *Next:* Lecture 8 builds the maps that preserve all this structure, the canonical maps, from a single function.

References

Faddeev and Yakubovskii, *Lectures on Quantum Mechanics for Mathematics Students*, §§2–3 (pp. 6–15): states as distributions, pure states, and the Jacobian proof of Liouville's theorem. Their bracket is the negative of (5.1), so their observable equation $\dot{f} = \{H, f\}$ and their density equation $\dot{\rho} = -\{H, \rho\}$ are our (5.2) and (7.7), and their $\rho(g_{-t}x)$ is our (7.6). Neumaier and Westra, *Classical and Quantum Mechanics via Lie Algebras*, §§1.7–1.8 (pp. 13–17): the Liouville equation as the adjoint of the Heisenberg equation. Arnold, *Mathematical Methods of Classical Mechanics*, §16 (pp. 68–70): Liouville's theorem by the divergence.